Emerging Directions in Deep Learning

Lab 1 - *Geomstats* - Introduction

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The aim of the lab is to provide a brief introduction to the *Geomstats* package, see also the GitHub repository. The reference paper is Miolane et al., 2020, see also this paper. The tasks of the lab can be found below. You will send a single PDF file including screenshots of the solutions and brief explanations where required on the individual chat of MSTeams. Deadline: **Thursday, 12.03.2026, 12:00**.

1. **(4p)** Install the *Geomstats* package and run / modify the notebooks `L_01_*` provided in the Notebooks folder.
   (a) Test various examples of manifolds ($\mathbb{R}^2, \mathbb{R}^3, S^2$) by changing the notebook `L_01_1`.
   (b) Explore the data set `cities`. How many cities are in this data set?
   (c) Explore the data set `poses`.
   (d) Explore (i) plane vectors; (ii) vectors having the origin on the sphere; (iii) geodesic on the sphere (induced by an origin and a direction vector); (iv) geodesic on the sphere (determined by two points); (v) geodesics in the hyperbolic plane - the Poincaré model.

2. **(2p)** One considers the points $A = (0, 1, 0)$ şi $B = (0, 0, 1)$ on the sphere $S^2 \subset \mathbb{R}^3$ having as center the origin $O = (0, 0, 0)$ and the radius 1. Compute the geodesic distance on the sphere between the points A and B (search keyword: `metric`).

3. **(2p)** One considers the symmetric positive definite matrices
$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$
Compute the distance between the matrices as elements of the SPD(2) manifold, endowed with the affine-invariant metric.

   Remark. For two matrices $X, Y \in$ SPD(2), the distance given by the affine-invariant metric is
$$d(X, Y) = N(X^{-\frac{1}{2}} Y X^{-\frac{1}{2}}),$$
where (i) $X^{\frac{1}{2}}$ is that matrix Z such that $Z^2 = X$, (ii) for a symmetric positive definite matrix S with eigenvalues $\lambda_1, \ldots, \lambda_p$ one considers
$$N(S)^2 = \sum_{i=1}^{p} (\ln(\lambda_i))^2.$$

4. **(2p)** Run and briefly present the outcomes of one of the following tutorials from the *Geomstats* site:
   (a) Learning on tangent data.
   (b) Fréchet mean and tangent PCA.
   (c) K-Means clustering on a Riemannian Manifold